

2 Student Grade Calculator :

```
import java.util.Scanner;
import java.util.Locale;

public class GradeCalculator {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);
        scanner.useLocale(Locale.US); // For consistent decimal formatting

        int numSubjects = getNumberOfSubjects(scanner);
        String[] subjectNames = getSubjectNames(scanner, numSubjects);
        int[] marks = getMarks(scanner, subjectNames);

        int totalMarks = calculateTotal(marks);
        double averagePercentage = calculateAveragePercentage(totalMarks,
numSubjects);
        char grade = calculateGrade(averagePercentage);
        int highestMark = findHighest(marks);
        int lowestMark = findLowest(marks);

        printReport(subjectNames, marks, totalMarks, averagePercentage, grade,
highestMark, lowestMark);

        scanner.close();
    }
}
```

```

private static int getNumberOfSubjects(Scanner scanner) {
    int numSubjects;
    do {
        System.out.print("Enter the number of subjects: ");
        while (!scanner.hasNextInt()) {
            System.out.print("Invalid input. Enter a positive integer for number of
subjects: ");
            scanner.next();
        }
        numSubjects = scanner.nextInt();
    } while (numSubjects <= 0);
    return numSubjects;
}

```

```

private static String[] getSubjectNames(Scanner scanner, int numSubjects) {
    String[] names = new String[numSubjects];
    scanner.nextLine(); // consume leftover newline
    for (int i = 0; i < numSubjects; i++) {
        System.out.print("Enter name of subject " + (i + 1) + ": ");
        names[i] = scanner.nextLine().trim();
        if (names[i].isEmpty()) {
            names[i] = "Subject " + (i + 1);
        }
    }
    return names;
}

```

```
private static int[] getMarks(Scanner scanner, String[] subjectNames) {  
    int[] marks = new int[subjectNames.length];  
    for (int i = 0; i < subjectNames.length; i++) {  
        marks[i] = getValidMark(scanner, subjectNames[i]);  
    }  
    return marks;  
}
```

```
private static int getValidMark(Scanner scanner, String subjectName) {  
    int mark;  
    do {  
        System.out.print("Enter marks obtained in " + subjectName + " (0-100):  
");  
        while (!scanner.hasNextInt()) {  
            System.out.print("Invalid input. Enter an integer between 0 and 100:  
");  
            scanner.next();  
        }  
        mark = scanner.nextInt();  
    } while (mark < 0 || mark > 100);  
    return mark;  
}
```

```
private static int calculateTotal(int[] marks) {  
    int total = 0;  
    for (int mark : marks) {
```

```
        total += mark;
    }
    return total;
}
```

```
private static double calculateAveragePercentage(int totalMarks, int
numSubjects) {
    return (double) totalMarks / (numSubjects * 100) * 100;
}
```

```
private static char calculateGrade(double averagePercentage) {
    if (averagePercentage >= 90) {
        return 'A';
    } else if (averagePercentage >= 80) {
        return 'B';
    } else if (averagePercentage >= 70) {
        return 'C';
    } else if (averagePercentage >= 60) {
        return 'D';
    } else {
        return 'F';
    }
}
```

```
private static int findHighest(int[] marks) {
    int highest = marks[0];
    for (int mark : marks) {
```

```

        if (mark > highest) {
            highest = mark;
        }
    }
    return highest;
}

```

```

private static int findLowest(int[] marks) {
    int lowest = marks[0];
    for (int mark : marks) {
        if (mark < lowest) {
            lowest = mark;
        }
    }
    return lowest;
}

```

```

private static void printReport(String[] subjectNames, int[] marks, int
totalMarks, double averagePercentage, char grade, int highestMark, int
lowestMark) {
    System.out.println("\n----- Grade Report -----");
    for (int i = 0; i < subjectNames.length; i++) {
        double subjectPercentage = (double) marks[i];
        System.out.printf("%-15s: %3d / 100 (%.2f%%)%n", subjectNames[i],
marks[i], subjectPercentage);
    }
    System.out.println("-----");
}

```

```
        System.out.println("Total Marks      : " + totalMarks + " / " +  
(subjectNames.length * 100));  
  
        System.out.printf("Average Percentage: %.2f%%\n", averagePercentage);  
  
        System.out.println("Grade          : " + grade);  
  
        System.out.println("Highest Mark    : " + highestMark);  
  
        System.out.println("Lowest Mark     : " + lowestMark);  
    }  
}
```